# Multi-Aperture Transformers for 3D (MAT3D) Segmentation of Clinical and Microscopic Images

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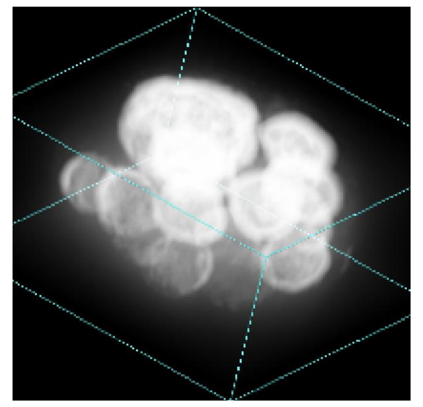
## Background:

### Motivation:

- Designing a scalable robust framework for 3D clinical and microscopy segmentation.
- Fuse the local and global dependencies using convolutional and transformer blocks.
- Enhance the loss function by integrating morphometric properties.

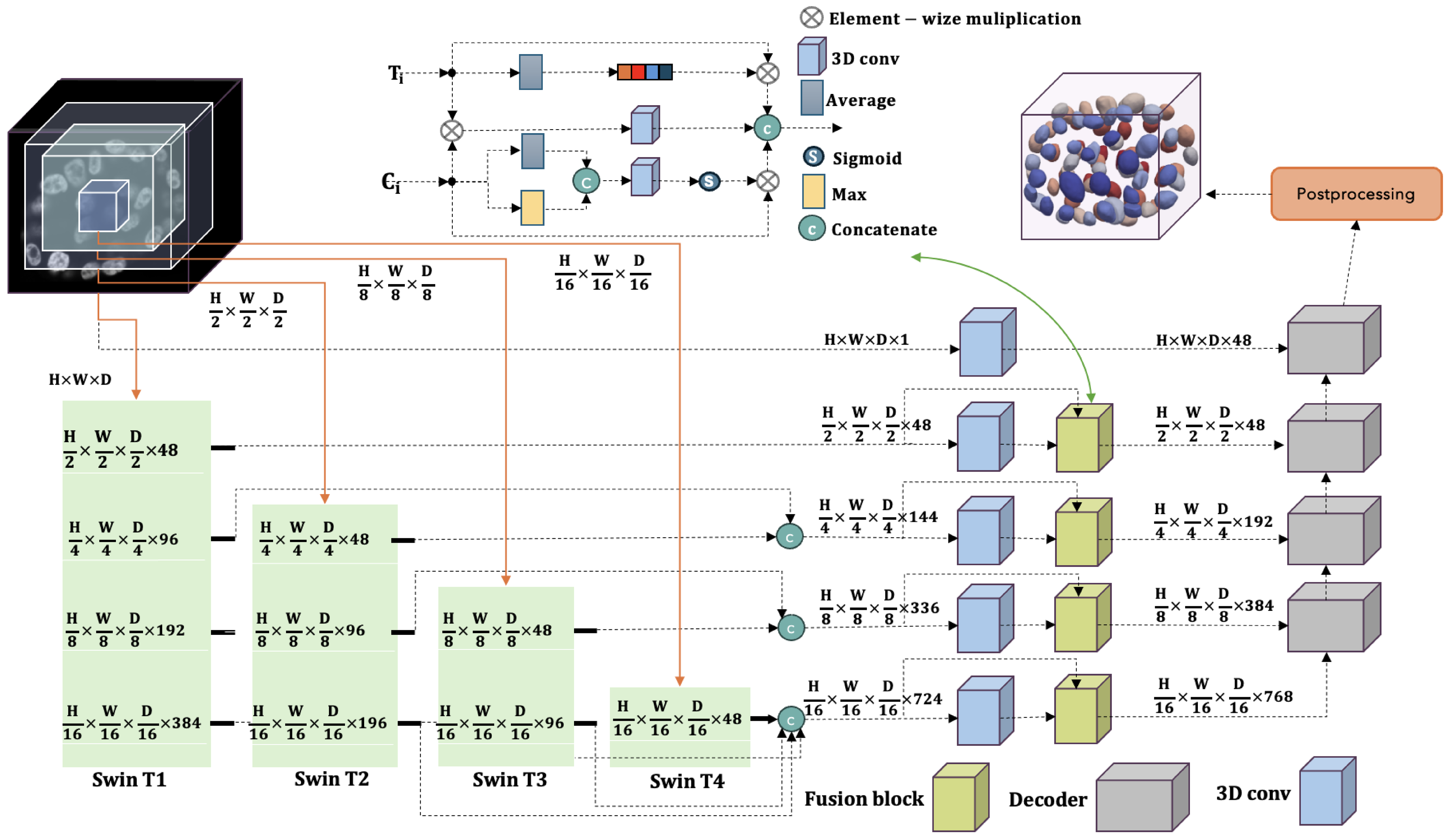


3D Organoid Representation

### Contributions:

- Four parallel Swin Transformers that couple the pyramid representation with the original resolution, i.e. multi-aperture representation.
- A 3D fusion block with a squeeze and excitation block and a convolutional block attention module (CBAM) for combining the outputs of each Transformer and its convolutional block.
- A loss function that integrates the soft Dice loss, with cross-entropy loss, the number of cells, and the distance transform.
- Tested on Synapse, Microscopy, and ACDC for robustness.
- A reduce model complexity with 40M parameters—an improvement over published result.

## Framework:




## Ablation study:

| Loss Function | Dice↑ | PQ↑ |
|---|---|---|
| $L_{dice}$ | 94.14 | 95.10 |
| $L_{focal}$ | 94.31 | 95.31 |
| $L_{dice} + L_{ce}$ | 94.40 | 95.81 |
| $L_{dice} + L_{ce} + L_{CellCount}$ | 95.07 | 96.14 |
| $L_{dice} + L_{ce} + L_{DistLoss}$ | 95.08 | 96.58 |
| **$L_{dice}+L_{ce}+L_{CellCount}+L_{DistLoss}$** | **95.12** | **97.01** |

Effects of the alternative Loss function

| Patch Size | Dice↑ | PQ↑ |
|---|---|---|
| [16, 16, 16] | 92.90 | 93.51 |
| [32, 32, 32] | 93.31 | 94.10 |
| [48, 48, 48] | 93.95 | 94.80 |
| [64, 64, 64] | 94.61 | 95.83 |
| **[96, 96, 96]** | **95.12** | **97.01** |
| [128, 128, 128] | 93.70 | 94.10 |

Hyperparameter tuning for the patch size

| Ratio(%) | SAM | | 3D U-NET | | 3DTransUNet | | Swin UNETR | | MAT3D | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Dice↑ | PQ↑ | Dice↑ | PQ↑ | Dice↑ | PQ↑ | Dice↑ | PQ↑ | Dice↑ | PQ↑ |
| 5 | 51.03 | 46.05 | 56.78 | 52.34 | 78.10 | 76.55 | 85.41 | 83.22 | 91.63 | 92.30 |
| 10 | 61.75 | 56.89 | 67.23 | 62.54 | 81.80 | 79.41 | 85.79 | 83.87 | 92.34 | 93.35 |
| 40 | 71.24 | 65.23 | 75.62 | 72.13 | 82.31 | 80.90 | 86.14 | 84.52 | 93.47 | 94.18 |
| 60 | 75.43 | 68.34 | 78.45 | 74.36 | 85.81 | 85.02 | 88.37 | 85.45 | 94.23 | 95.24 |
| **80** | **77.73** | **70.26** | **80.65** | **79.34** | **86.46** | **85.09** | **89.74** | **86.00** | **95.12** | **97.01** |

MAT3D converges faster with the smaller training size

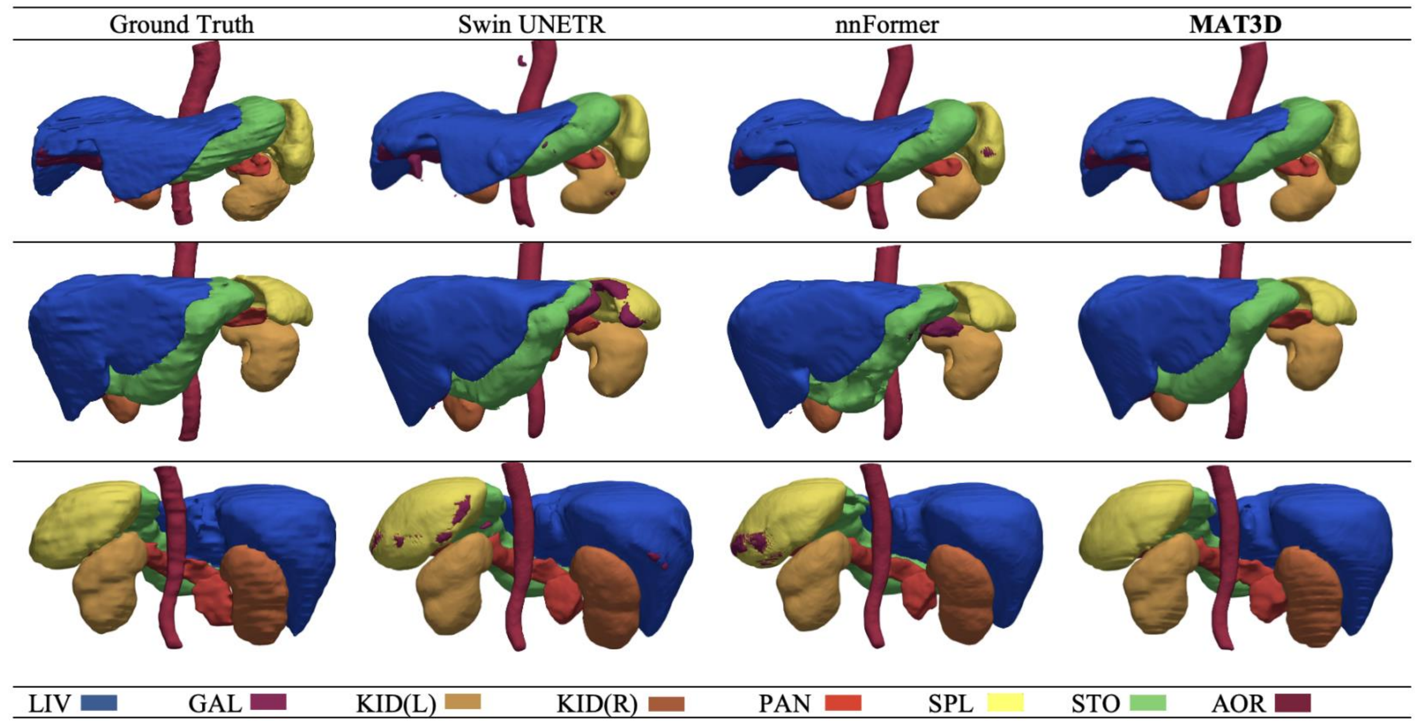
| Blocks | Dice↑ | PQ↑ | Parameters |
|---|---|---|---|
| $T_1$ | 92.10 | 93.01 | 26M |
| $T_1 + T_2$ | 92.65 | 93.30 | 30M |
| $T_1 + T_2 + T_3$ | 93.40 | 94.97 | 34M |
| $T_1 + T_2 + T_3 + T_4$ | 94.02 | 95.91 | 36M |
| **$T_1 + .. + T_4$ + Fusion** | **95.12** | **97.01** | **40M** |

Increasing number of transformers and Fusion Blocks improve performance

## Results: Comparable performance with SOTA

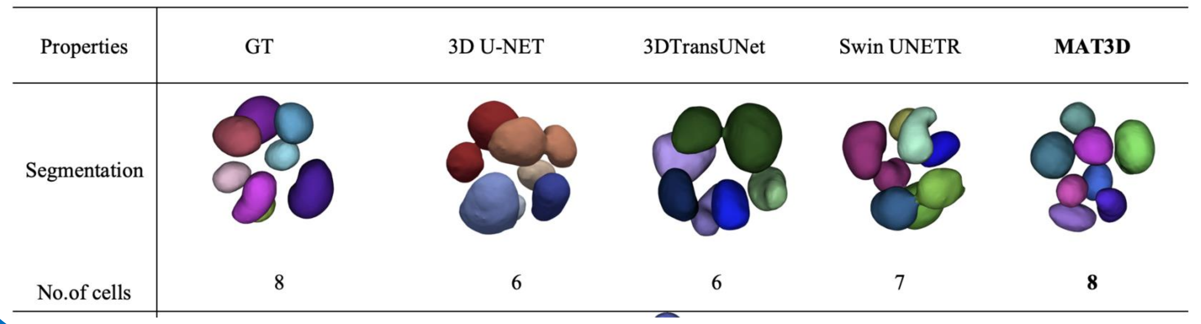
### Synapse Data

| Methods | Average | | | Spl | Kid(R) | Kid(L) | Gal | Liv | Sto | Aor | Pan |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Dice↑ | HD95↓ | Parameters | | | | | | | | |
| MIST | 86.92 | 11.07 | - | 92.83 | 93.28 | 92.54 | 74.58 | 94.94 | **87.23** | 89.15 | 72.43 |
| Swin UNETR | 83.47 | 10.55 | 62M | 95.37 | 86.26 | 86.99 | 66.54 | 95.72 | 77.01 | 91.12 | 68.80 |
| nnFormer | 86.57 | 10.63 | 150M | 90.51 | 86.25 | 86.57 | 70.17 | 96.84 | 86.83 | 92.04 | **83.35** |
| UNETR++ | 87.22 | 07.53 | 42M | 95.77 | 87.18 | 87.54 | 71.25 | 96.42 | 86.01 | 92.52 | 81.10 |
| MISSFormer | 81.96 | 18.20 | - | 91.92 | 82.00 | 85.21 | 68.65 | 94.41 | 80.81 | 86.99 | 65.67 |
| LeVit-UNet | 78.53 | 16.84 | - | 88.86 | 80.25 | 84.61 | 62.23 | 93.11 | 72.76 | 87.33 | 59.07 |
| 3DTransUNet | 88.10 | - | 81M | 93.39 | 87.47 | 85.76 | **81.15** | 97.03 | 85.31 | 92.97 | 81.76 |
| MAT3D | **89.73** | **07.31** | 40M | **96.51** | **93.35** | **93.64** | 76.87 | **97.16** | 85.69 | **93.05** | 81.20 |




### Microscopy Data

| Method | Dice↑ | PQ↑ | Parameters |
|---|---|---|---|
| OTSU Thresholding | 64.67 | 59.76 | N/A |
| SAM 2D | 77.73 | 70.26 | 91M |
| 3D U-NET | 80.65 | 79.34 | 95M |
| 3DTransUNet | 86.46 | 85.09 | 81M |
| Swin UNETR | 89.74 | 86.00 | 61M |
| 3D-Organoid-Swin | 94.91 | – | 21M |
| **MAT3D** | **95.12** | **97.01** | **40M** |




### ACDC Data

| Methods | Dice↑ | RV | Myo | LV |
|---|---|---|---|---|
| VIT-CUP | 81.45 | 81.46 | 70.71 | 92.18 |
| R50-VIT-CUP | 87.57 | 86.07 | 81.88 | 94.75 |
| UNETR | 88.61 | 85.29 | 86.52 | 94.02 |
| LeViT-UNet | 90.32 | 89.55 | 87.64 | 93.76 |
| SegFormer3D | 90.96 | 88.50 | 88.86 | 95.53 |
| nnFormer | 92.06 | 90.94 | 89.58 | 95.65 |
| MIST | 92.56 | 91.23 | 90.31 | 96.14 |
| UNETR++ | 92.83 | 91.89 | 90.61 | 96.00 |
| **MAT3D** | **93.34** | **92.55** | **91.06** | **96.41** |